

## LESSON 5-5 • SESSION 1 • PRACTICE • FORM B

**Volume of Rectangular Prisms**

Learning target – I can find the volume of a rectangular prism by multiplying length, width, and height. · 6.G.A.2

**REMEMBER**

- **Volume** is the space inside a solid, counted in unit cubes. Write **cubic units**:  $\text{cm}^3$ ,  $\text{in}^3$ ,  $\text{ft}^3$ .
- One layer = length  $\times$  width = **base area B**. The height counts the layers.
- **$V = l \times w \times h$  or  $V = B \times h$ .**

**WORKED EXAMPLE**

A box is 6 cm long, 3 cm wide, and 4 cm tall.

**Step 1** One layer:  $6 \times 3 = 18$  cubes.**Step 2** 4 layers stack up.**Step 3**  $18 \times 4 = 72$ .**Answer  $V = 72 \text{ cm}^3$** **WORDS** **volume** (*volumen*) · **rectangular prism** (*prisma rectangular*) · **cubic units** (*unidades cúbicas*)**PRACTICE***show your work · label units*

- 1.**
- Find the volume of each rectangular prism.

**a.**  $9 \text{ cm} \times 5 \text{ cm} \times 6 \text{ cm}$  \_\_\_\_  $\text{cm}^3$

**b.**  $8 \text{ ft} \times 3 \text{ ft} \times 2 \text{ ft}$  \_\_\_\_  $\text{ft}^3$

**c.**  $10 \text{ ft} \times 7 \text{ ft} \times 2 \text{ ft}$  \_\_\_\_  $\text{ft}^3$

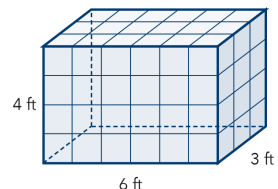
**d.**  $B = 36 \text{ in}^2$ ,  $h = 8 \text{ in}$  \_\_\_\_  $\text{in}^3$

- 2. Toy Chest.**
- A toy chest is packed with cubes 1 ft on each edge, as shown.

**a.** Cubes in the bottom layer \_\_\_\_\_

**b.** Number of layers \_\_\_\_\_

**c.** Volume of the chest \_\_\_\_  $\text{ft}^3$

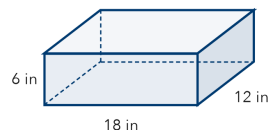


- 3. Fish Tank.**
- A fish tank is 18 in long, 12 in wide, and 6 in tall.

**Hint** Find the base first. The height counts the layers.

**a.** Area of the base \_\_\_\_  $\text{in}^2$

**b.** Volume of the tank \_\_\_\_  $\text{in}^3$

*Show your work.*

- 4. Locker Space.**
- A locker is 14 in wide, 12 in deep, and 50 in tall. Find its volume.

**Hint** Base = width  $\times$  depth. Then multiply by the height.*Show your work.***Answer** \_\_\_\_\_  $\text{in}^3$ 

- 5. Sandbox Fill.**
- A sandbox is 24 ft long and 12 ft wide. Sand is poured in 2 ft deep. How much sand is needed?

**Hint** The depth of the sand is the height.*Show your work.***Answer** \_\_\_\_\_  $\text{ft}^3$

6. **Rosa's Box.** A box is 19 cm long, 11 cm wide, and 4 cm tall. Rosa says its volume is  $209 \text{ cm}^3$ . Is Rosa right?

**Hint** How many edges did Rosa use? A prism has three.

Show your work.

**True volume** \_\_\_\_\_

7. **Two Cartons.** Carton A is 16 cm by 6 cm by 10 cm. Carton B is 12 cm by 10 cm by 7 cm. Which carton holds more? How much more?

Show your work.

**Carton, how much** \_\_\_\_\_

8. **Aquarium.** An aquarium is 50 cm long, 30 cm wide, and 40 cm tall. One liter of water fills  $1,000 \text{ cm}^3$ .

- a. Find the volume. \_\_\_\_\_  $\text{cm}^3$
- b. How many liters fill it? \_\_\_\_\_ L
- c. It is filled only 30 cm deep. How many liters? \_\_\_\_\_ L

Show your work.

## FIND THE ERROR

explain the mistake, then fix it

9. **Malik's Work.** A crate is 12 ft long, 9 ft wide, and 5 ft tall. Malik wrote: " $V = 12 + 9 + 5 = 26 \text{ ft}^3$ ." Explain the mistake and find the correct volume.

---



---



---

## TEST PRACTICE

choose the best answer

10. A prism is 2 m by 6 m by 5 m. Its volume is –
- A**  $13 \text{ m}^3$       **B**  $60 \text{ m}^2$       **C**  $60 \text{ m}^3$       **D**  $70 \text{ m}^3$
11. In  $V = B \times h$ , the letter B stands for the –
- A** base area      **B** biggest edge      **C** box      **D** base edge

## SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

|         |         |          |            |          |         |          |          |            |          |
|---------|---------|----------|------------|----------|---------|----------|----------|------------|----------|
| 72<br>C | 60<br>N | 836<br>A | 8,400<br>N | 209<br>Q | 45<br>S | 576<br>T | 120<br>I | 1,296<br>O | 168<br>Z |
| #2c     | #3b     | #4       | #5         | #6       | #7      | #8b      | #8c      |            |          |

## CHECK YOURSELF

☐ I can do this on my own    ☐ I need a little help    ☐ I have a question about #\_\_\_\_\_